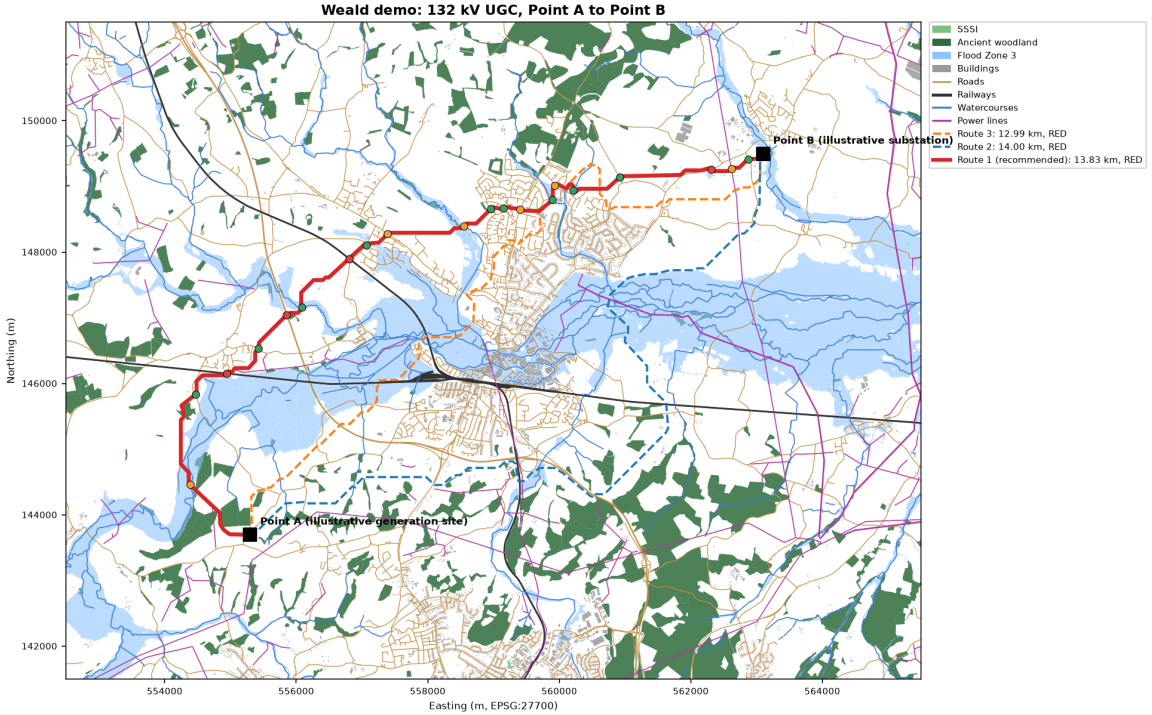
# RAG Risk & Crossings Report: Weald demo: 132 kV UGC, Point A to Point B

Illustrative 132 kV underground cable corridor study, Tonbridge area, Kent.
Point A (illustrative generation site) to Point B (illustrative substation). Generated 2026-11-09. Coordinates: British National Grid (EPSG:27700).

## *Route comparison*

| Route | Recommended | Length (m) | Weighted cost | Crossings | By type | RED | AMBER | RAG |
|---|---|---|---|---|---|---|---|---|
| 1 | Yes | 13833.0 | 118841.0 | 20 | Power line: 1, Railway: 2, Road: 13, Watercourse: 4 | 6 | 26 | RED |
| 2 | | 13996.0 | 130032.0 | 18 | Power line: 3, Railway: 2, Road: 7, Watercourse: 6 | 6 | 17 | RED |
| 3 | | 12991.0 | 130213.0 | 28 | Power line: 1, Railway: 2, Road: 17, Watercourse: 8 | 5 | 26 | RED |



## *Crossings: recommended route (13.83 km)*

| Ref | Chainage | Type | Class | Name / ref | Angle | RAG | Reason |
|---|---|---|---|---|---|---|---|
| X01 | 1+328 | Watercourse | river | River Medway | 51 deg | AMBER | Watercourse: river; oblique crossing (51 deg) |
| X02 | 2+881 | Road | tertiary | - | 87 deg | GREEN | Road: tertiary |
| X03 | 3+616 | Railway | rail | Redhill to Tonbridge Line | 84 deg | RED | Railway: rail |
| X04 | 4+361 | Road | unclassified | - | 71 deg | GREEN | Road: unclassified |
| X05 | 5+043 | Road | trunk | A21 | 76 deg | RED | Road: trunk; parallel features over 13 m (e.g. dual carriageway) |
| X06 | 5+346 | Watercourse | stream | - | 78 deg | GREEN | Watercourse: stream |
| X07 | 6+503 | Railway | rail | South Eastern Main Line | 81 deg | RED | Railway: rail |
| X08 | 6+859 | Road | residential | - | 89 deg | GREEN | Road: residential |
| X09 | 7+260 | Road | secondary | B245 | 78 deg | AMBER | Road: secondary |
| X10 | 8+464 | Watercourse | stream | Hilden Brook | 47 deg | AMBER | Watercourse: stream; oblique crossing (47 deg) |
| X11 | 8+983 | Road | residential | - | 81 deg | GREEN | Road: residential |
| X12 | 9+194 | Road | residential | - | 87 deg | GREEN | Road: residential |
| X13 | 9+472 | Road | primary | A227 | 82 deg | AMBER | Road: primary |
| X14 | 10+024 | Watercourse | stream | Pen Stream | 72 deg | GREEN | Watercourse: stream |
| X15 | 10+277 | Road | residential | - | 60 deg | AMBER | Road: residential; oblique crossing (60 deg) |
| X16 | 10+687 | Road | residential | - | 67 deg | GREEN | Road: residential |
| X17 | 11+483 | Road | unclassified | - | 79 deg | GREEN | Road: unclassified |
| X18 | 12+931 | Road | trunk | A26 | 81 deg | RED | Road: trunk |
| X19 | 13+256 | Power line | line | - | 65 deg | AMBER | Power line: line |
| X20 | 13+573 | Road | unclassified | - | 68 deg | GREEN | Road: unclassified |

## *Constraints, proximity, design and data checks: recommended route*

| Category | Item | Detail | From | To | RAG |
|---|---|---|---|---|---|
| area | Flood Zone 3 | 467 m of centreline inside | 1+231 | 1+698 | AMBER |
| area | Flood Zone 3 | 42 m of centreline inside | 3+558 | 3+600 | AMBER |
| area | Flood Zone 3 | 113 m of centreline inside | 5+325 | 5+438 | AMBER |

| Category | Item | Detail | From | To | RAG |
|---|---|---|---|---|---|
| area | Flood Zone 3 | 25 m of centreline inside | 6+911 | 6+936 | AMBER |
| area | Flood Zone 3 | 184 m of centreline inside | 8+302 | 8+486 | AMBER |
| area | Flood Zone 3 | 32 m of centreline inside | 10+015 | 10+054 | AMBER |
| area | Flood Zone 3 | 21 m of centreline inside | 10+457 | 10+478 | AMBER |
| area | Flood Zone 3 | 32 m of centreline inside | 10+622 | 10+692 | AMBER |
| area | Ancient woodland | construction swathe overlaps | 0+947 | 0+947 | RED |
| proximity | Within 15 m of ancient woodland (standing-advice buffer) | 5 feature(s) within 15 m; closest 4 m | 0+655 | - | RED |
| design | Sharp deflection | 73 deg change of direction | 2+612 | - | AMBER |
| design | Sharp deflection | 90 deg change of direction | 3+591 | - | AMBER |
| design | Sharp deflection | 90 deg change of direction | 3+641 | - | AMBER |
| design | Sharp deflection | 72 deg change of direction | 5+324 | - | AMBER |
| design | Sharp deflection | 80 deg change of direction | 6+834 | - | AMBER |
| design | Sharp deflection | 82 deg change of direction | 6+884 | - | AMBER |
| design | Sharp deflection | 81 deg change of direction | 10+247 | - | AMBER |
| design | Sharp deflection | 83 deg change of direction | 10+463 | - | AMBER |
| design | Sharp deflection | 78 deg change of direction | 10+682 | - | AMBER |
| design | Sharp deflection | 67 deg change of direction | 10+736 | - | AMBER |
| design | Sharp deflection | 63 deg change of direction | 12+909 | - | AMBER |
| design | Sharp deflection | 63 deg change of direction | 12+954 | - | AMBER |

## *Data sources*

| Layer | Status | Features | Attribution | Fetched (UTC) |
|---|---|---|---|---|
| sssi | ok | 4 | Contains Natural England data © Natural England copyright and database right | 2026-11-09T13:57:26+00:00 |
| ancient_woodland | ok | 378 | Contains Natural England data © Natural England copyright and database right | 2026-11-09T13:57:27+00:00 |
| flood_zone_3 | ok | 4176 | © Environment Agency copyright and/or database right. All rights reserved. | 2026-11-09T13:57:28+00:00 |
| buildings | ok | 9231 | © OpenStreetMap contributors, ODbL (production: OS MasterMap Topography) | 2026-11-09T13:57:37+00:00 |

| Layer | Status | Features | Attribution | Fetched (UTC) |
|---|---|---|---|---|
| roads | ok | 1967 | © OpenStreetMap contributors, ODbL (production: OS MasterMap Highways) | 2026-11-09T13:58:31+00:00 |
| railways | ok | 174 | © OpenStreetMap contributors, ODbL | 2026-11-09T13:59:59+00:00 |
| waterways | ok | 466 | © OpenStreetMap contributors, ODbL (production: OS / EA main rivers) | 2026-11-09T14:01:18+00:00 |
| power_lines | ok | 94 | © OpenStreetMap contributors, ODbL | 2026-11-09T14:02:21+00:00 |

**Limitations.** Desk-based screening only. Existing buried services (gas, water, telecoms, other electricity assets) are not in open data: a utility search and statutory undertaker enquiries are required. RAG rules are configurable defaults, not a substitute for engineering, environmental or planning judgement.